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INVITED TALK / ABSTRACT

Potentials for sub-Laplacians and geometric applications

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Abstract

In this talk I will report my recent joint research with Shiguang Ma. I will talk about the new approach to use the geometric completeness to estimate the Hausdorff dimension of polar sets of potentials of nonnegative Radon measures for sub-Laplacians in Carnot groups. In this new approach, one relies on inequalities that are analogous to the classic integral inequalities about Riesz kernels in Euclidean spaces. It also uses some geometric measure theoretic arguments and the polar coordinates by horizontal curves constructed by Balogh and Tyson for Carnot groups. As a consequence, we develop applications of potentials for sub-Laplacians in CR geometry, quaternionic CR geometry, and octonionic CR geometry.

Keywords: potentials; sub-Laplacians; Carnot groups; Hausdorff dimension; polar sets; CR geometry